Straight-line residual inspection
Mathematical and implementation guide, v2

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This guide describes the completed computation, ASDF and display workflow derived from the owner-approved design `docs/developer/design/linearity-residual-asdf-v2.md`. It operates on saved ROI means, not individual image pixels. It produces exploratory diagnostics, not physical shutter or amplifier calibrations.

1 Inputs and selected population

For one detector and ROI, each individual flat exposure supplies exposure time t_i in seconds and saved bias-subtracted mean signal S_i in ADU. Both members of each pair remain separate. Their common accepted pixel population, source asset/HDU, acquisition index and original table row remain attached. The input stage is explicit: legacy raw-pair signals have no overscan correction; preparation signals already have row-overscan and master-bias subtraction. Diagnostics and display perform no image correction. The reference-line population I contains finite (t_i, S_i) and satisfies the user's inclusive original-signal interval. Blank limits mean all available data. No weighting, outlier rejection or hidden exposure cut is applied. Points outside I remain visible and receive predictions if a line is available.

2 Centered ordinary least squares

With $n = |I|$, define

$$\bar{t} = \frac{1}{n}\sum_{i\in I} t_i, \qquad \bar{S} = \frac{1}{n}\sum_{i\in I} S_i,$$

$$b = \frac{\sum_{i\in I}(t_i - \bar{t})(S_i - \bar{S})}{\sum_{i\in I}(t_i - \bar{t})^2}, \qquad a = \bar{S} - b\bar{t}.$$

The slope b has units ADU/s and a has units ADU. A line requires at least two points and nonzero exposure-time spread. Invalid intervals, insufficient points, zero time spread and numerical overflow give explicit unavailable states. The computation service uses Python math.fsum for accurate summation. Means use the first sample as an origin, summing scaled differences; predictions use

$$\widehat{S}_i = \bar{S} + b(t_i - \bar{t})$$

rather than directly evaluating $a+bt_i$, reducing cancellation for large time origins. No coefficient uncertainties are inferred from these descriptive fits.

3 Residuals, rates and display

For each available original measurement, including outside-interval points,

$$r_i = S_i - \widehat{S}_i, \qquad r_{\%,i} = 100\frac{r_i}{\widehat{S}_i} \quad (\widehat{S}_i > 0), \qquad R_i = \frac{S_i}{t_i} \quad (t_i > 0).$$

Residuals in ADU remain available at nonpositive predictions, but percentage residuals do not. Zero-time count rates and unavailable signals remain null; negative measured signals are retained. Nonfinite derived values are null. Residual plots use original signal or exposure time on the horizontal axis, and percentage or ADU on the vertical axis, with a zero reference. Curves connect measurements for inspection; fitting limits affect the line, not displayed data. The per-ROI summary reports a, b, n, the actual selected time/signal

ranges and status. All fits, predictions, residuals, rates, settings and units are saved in ASDF before plotting. The HTML reads these values without recalculating them. Fit limits are read-only: a different interval requires a new computation run and a new immutable diagnostic product. The JSON download exports saved values and source references only. Original ASDF measurements are never rewritten.

4 What a shutter error can show

For an otherwise linear, correctly bias-subtracted response under stable light,

$$S_i = A(t_i + \delta) \quad \Longrightarrow \quad a = A\delta, \quad b = A, \quad R_i = A\left(1 + \frac{\delta}{t_i}\right).$$

A free-intercept line absorbs the constant shutter offset: residuals alone do not recover it. Its fractional count-rate effect grows as $1/t_i$, motivating the separate rate view. An additive residual bias also produces a $1/t_i$ term; lamp variation and detector nonlinearity can contribute structure. No physical shutter estimate is extracted here. Different signal intervals can yield different residual baselines even for the same underlying response.

5 Separate computation, persistence and plotting

Arithmetic: `pycavi_residuals.py`, function `analyze`, implements centered OLS and derived per-row values as a pure Python service. It does not read files, plot or change input samples.

Persistence: `pycavi_residual_qc.py` wraps the common QC factory, writer and reader. Each detector has an actual derived `line-residuals.tsv` transport and `line-residuals.asdf` sidecar. The named `line_residual_diagnostic` extension stores configuration, per-ROI fits, all samples/derived values, explicit units and the source pair digest. Original QC input graphs are copied locally with frozen semantic links; raw images are not copied or read. No common schema/writer modification is needed.

Computation command: `tools/compute-pair-residuals.py` takes a completed pair-ASDF batch (or one pair product), explicit fit limits and a fresh output directory. It writes one diagnostic ASDF per detector and a completion index. Exact round-trips and input-preservation checks must pass. A failed run remains incomplete and is never automatically overwritten or resumed.

Plotting command: `tools/view-pair-qc.py` accepts the original batch plus a completed diagnostic index via `--diagnostics`. The reader validates source and output record digests without repeating the fit. The HTML embeds only saved data. Axis, detector, ROI and unit changes are presentation operations; there is no regression, prediction, residual or rate arithmetic in the browser. Without diagnostic products, original measurement plots still work and missing diagnostics are explicit.

Validation: `test_residual_qc.py` checks analytical solutions, restricted intervals, null/signed/zero cases, independent ROIs, rank failure, large time origins, ASDF round-trip/relocation, frozen lineage and computation without plotting. Reader and viewer tests disable the fitter to prove that plotting cannot refit. Viewer JavaScript checks plot supplied stored values verbatim. Separate Mac and browser acceptance remain necessary.

The former browser-side computation and v1 guide are historical. ASDF is the numerical result; TSV, JSON and HTML are supporting transport or inspection products. No physical correction is adopted by this diagnostic.

6 Corrected preparation bridge (5 November)

`pycavi_measurement_display.read_source` also accepts immutable preparation ASDF. It projects the saved individual exposure/ROI means and spatial variances without averaging pair members or loading raw images/native files. Asset, HDU, input index, pair/member, ROI coordinates and common pixel population remain explicit. ROI means stay distinct from direct-pixel samples.

The existing `pycavi_residuals.analyze` equations above are unchanged. Residual ASDF freezes the full source graph and carries the actual stage and source kind. The generic source-measurement digest is authoritative; the old source-pair digest key is a compatibility alias for that same record. Readers check stage/kind against the source, preserve all original samples and consume stored curves without fitting. Later ratio products inherit that lineage.

The existing HTML/axis/cursor component renders corrected signal/time, spatial variance, residuals, rates and ratios. Its gain and bias-RON channels retain their separately labelled raw-pair meanings; spatial scatter is not an assigned uncertainty of a mean. Mixed corrected/raw source stages are refused within one mosaic. Changing a diagnostic interval creates new ASDF products; the original preparation and older diagnostic archives remain immutable.